

Subject Code: BAS-101

Time : 1 ½ Hours

Subject: APPLIED MATHEMATICS-1

Maximum Marks : 30

Q1 Attempt all the parts

(a) For which value of 'b' the rank of the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ b & 13 & 10 \end{bmatrix}$ is 2. (2.5*4)

(b) Express $\begin{bmatrix} -1 & 7 & 1 \\ 2 & 3 & 4 \\ 5 & 0 & 5 \end{bmatrix}$ matrix as the sum of a symmetric and a skew symmetric matrix .

(c) Test for convergence of the series $\sum (1 + \frac{1}{\sqrt{n}})^{-n^{3/2}}$.

(d) Examine the convergence of the series

$$\sum \frac{(n+1)(n+2)}{(n^2+1)(n^2+2)}$$

Q2 (Attempt any Two Parts) UNIT-1

(5,5)

(a) Test for the consistency of the following system of linear equations and hence find the solution

$$\begin{aligned} 4x_1 - x_2 &= 12 \\ -x_1 + 5x_2 - 2x_3 &= 0 \\ -2x_2 + 4x_3 &= -8 \end{aligned}$$

(b) Let $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$, find its modal matrix P such that $P^{-1}AP$ is diagonal matrix.

(c) Find the inverse of the matrix by employing the elementary transformation $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

Q3 (Attempt any Two Parts) UNIT-2

(5,5)

(a) Test the convergence of the series ($x > 0$)

$$\frac{1}{1.2}x + \frac{1}{2.3}x^2 + \frac{1}{3.4}x^3 + \dots$$

(b) Examine the series $\sum \frac{(-1)^n}{n(\log n)^2}$ for convergence and absolute convergence

(c) Show that the positive term geometric series $1 + r + r^2 + r^3 + r^4 + \dots$ converges for $r < 1$ and diverges to ∞ for $r \geq 1$.